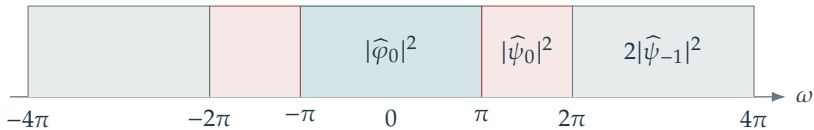
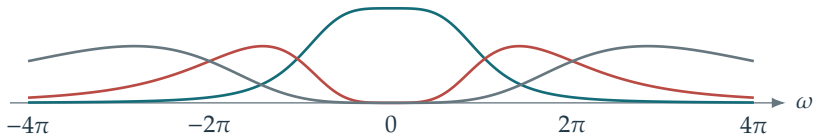


Shannon: sharp bands



Splines: smooth overlap (schematic)



$$2^{-j} |\hat{\psi}_{j,0}(\omega)|^2 = |\hat{\psi}(2^j \omega)|^2$$

$$\sum_{j \in \mathbb{Z}} |\hat{\psi}(2^j \omega)|^2 = 1 \quad (\text{a.e. } \omega \neq 0)$$